Deniz Ertuğrul

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[Linkedin Profile] • [Github Profile]

Skills

- **Programming & Data Science**: **Python** (PyTorch, Scikit-learn, Pandas), **R** (Bioconductor, Tidyverse), **SQL**, **Bash**.
- **Bioinformatics**: **Single-Cell RNA-seq** (scRNA-seq), **Transcriptomics** (RNA-seq), **Epigenomics** (DNA Methylation).
- **Advanced Computational Methods: Molecular Dynamics** (GROMACS, CHARMM), **Virtual Screening**, Weighted Gene Co-expression Network Analysis (**WGCNA**), Meta-Analysis, **Machine Learning**, and **Deep Learning**

PROFESSIONAL EXPERIENCE

CANCER SIGNALING LABORATORY — **BOĞAZIÇI UNIVERSITY**

Project Intern — **Dec 2023 – June 2024**

- Led **virtual screening** of 120,000 ligands to identify **GPCR 26** drug candidates.
- Conducted **molecular dynamics simulations** up to **20-ns** to validate ligand stability.
- **Optimized lead compounds** to enhance drug specificity.
- Utilized **Schrödinger Maestro, AlphaFold, GROMACS, and CHARMM** for computational analysis.

OGÜN LABORATORY — **SABANCI UNIVERSITY**

Project Intern — **Oct 2022 - May 2023**

- Analyzed **human single amino acid variations** to identify **disease-related mutations** using an evolution-based scoring system **PHACT model (Kuru et al., 2022).**
- Built a **Python-based machine learning model** to predict mutation pathogenicity.

PROJECTS

Pulmonary Arterial Hypertension (PAH) Gene Expression Analysis

- Integrated bioinformatics analysis of Pulmonary Arterial Hypertension (**PAH**) datasets to identify **311 differentially expressed genes (DEGs)**, **10 key hub genes** (e.g., ***POSTN***, ***SPP1***), and **enriched inflammatory pathways**.
- Immune cell infiltration analysis revealed **altered levels of monocytes, neutrophils, and macrophages**; predicted potential PAH therapeutics via Comparative Toxicogenomic Database.

HCC Meta-Analysis Pipeline

- Developed a pipeline for meta-analysis of differentially expressed genes in **Hepatocellular Carcinoma** (HCC) across 8 gene expression datasets.
- Identified **83** consistently **dysregulated genes**. Functional analysis revealed **up-regulated** pathways were linked to **cell proliferation**, while **down-regulated** genes were associated with **metabolic processes**.

EDUCATION

UNIVERSITY OF BOLOGNA — **BOLOGNA, ITALY**

MASTER OF SCIENCE, BIOINFORMATICS — **2024- PRESENT**

BOĞAZİÇİ UNIVERSITY — **Istanbul, Türkiye**

Bachelor of Science, Molecular Biology and Genetics — **2018-2024**

ADDITIONAL INFORMATION

- Advanced proficiency in English
- Professional musician with strong skills in improvisation, composition, and production. Effectively balanced music, academics, and research, demonstrating discipline and time management.